

MTH165: Mathematics for Engineers

Master Formula & Rapid Revision Book

Complete Theorem Checklists, Standard Derivative/Integral Tables & High-Yield Memorization Sheets

B.Tech / B.E. Semester 1 Curriculum | Lovely Professional University

LPU Engineering Open Publishing Initiative

Independent Open-Source Academic Guide Series

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Chapter 1

Unit 1: Matrix Methods and Linear Systems

Matrix Rank & Determinantal Minor Laws

- **Rank Definition:** $\text{rank}(A) = \rho(A)$ = Number of non-zero rows in Row Echelon Form (REF).
- If A is $m \times n$, then $\rho(A) \leq \min(m, n)$.
- $\rho(AB) \leq \min(\rho(A), \rho(B))$.
- $\rho(A + B) \leq \rho(A) + \rho(B)$.
- $\rho(A^T) = \rho(A) = \rho(A^T A) = \rho(AA^T)$.

Linear Systems Solvability ($Ax = b$)

Let $\tilde{A} = [A|b]$ be the augmented matrix for an $m \times n$ system:

1. **Inconsistent (No Solution):** $\rho(A) < \rho([A|b])$.
2. **Consistent (Unique Solution):** $\rho(A) = \rho([A|b]) = n$ (number of unknowns).
3. **Consistent (Infinite Solutions):** $\rho(A) = \rho([A|b]) = r < n$, with $(n - r)$ free independent parameters.

For homogeneous systems $Ax = 0$:

- Non-trivial solutions exist $\iff \rho(A) < n \iff \det(A) = 0$ (for square matrix).

Eigenvalues & Cayley–Hamilton Formulae

- **Characteristic Equation (3×3):**

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0 \quad (1.1)$$

where $S_1 = \text{tr}(A)$, $S_2 = M_{11} + M_{22} + M_{33}$, and $S_3 = \det(A)$.

- **Sum of Eigenvalues:** $\sum \lambda_i = \text{tr}(A) = a_{11} + a_{22} + a_{33}$.
- **Product of Eigenvalues:** $\prod \lambda_i = \det(A)$.
- **Cayley–Hamilton Theorem:** Every square matrix satisfies its characteristic polynomial:

$$A^3 - S_1A^2 + S_2A - S_3I = O \quad (1.2)$$

- **Matrix Inverse:** $A^{-1} = \frac{1}{S_3}(A^2 - S_1A + S_2I)$.

Chapter 2

Unit 2: Differential Calculus and Applications

Standard n -th Derivatives

- $\frac{d^n}{dx^n} [e^{ax}] = a^n e^{ax}$
- $\frac{d^n}{dx^n} [(ax + b)^m] = \frac{m!}{(m-n)!} a^n (ax + b)^{m-n}$
- $\frac{d^n}{dx^n} \left[\frac{1}{ax+b} \right] = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$
- $\frac{d^n}{dx^n} [\ln(ax + b)] = \frac{(-1)^{n-1} (n-1)! a^n}{(ax+b)^n}$
- $\frac{d^n}{dx^n} [\sin(ax + b)] = a^n \sin \left(ax + b + \frac{n\pi}{2} \right)$
- $\frac{d^n}{dx^n} [\cos(ax + b)] = a^n \cos \left(ax + b + \frac{n\pi}{2} \right)$
- $\frac{d^n}{dx^n} [e^{ax} \sin(bx + c)] = (a^2 + b^2)^{n/2} e^{ax} \sin \left(bx + c + n \tan^{-1} \frac{b}{a} \right)$

Leibniz's Product Rule

$$(uv)_n = \sum_{k=0}^n \binom{n}{k} u_{n-k} v_k = u_n v + \binom{n}{1} u_{n-1} v_1 + \binom{n}{2} u_{n-2} v_2 + \cdots + u v_n \quad (2.1)$$

Mean Value Theorems & Series Expansions

- **Rolle's:** $f(a) = f(b) \implies f'(c) = 0$ for some $c \in (a, b)$.
- **LMVT:** $f'(c) = \frac{f(b)-f(a)}{b-a}$ for some $c \in (a, b)$.
- **CMVT:** $\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$ for some $c \in (a, b)$.
- **Taylor's Series:** $f(a + h) = \sum_{k=0}^n \frac{h^k}{k!} f^{(k)}(a) + \frac{h^{n+1}}{(n+1)!} f^{(n+1)}(a + \theta h)$.

Chapter 3

Unit 3: Fundamentals of Integral Calculus

Master Properties of Definite Integrals

- $\int_a^b f(x) dx = \int_a^b f(a + b - x) dx$ (**King's Property**)
- $\int_0^a f(x) dx = \int_0^a f(a - x) dx$
- $\int_{-a}^a f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & f(-x) = f(x) \text{ (Even)} \\ 0, & f(-x) = -f(x) \text{ (Odd)} \end{cases}$
- $\int_0^{2a} f(x) dx = \begin{cases} 2 \int_0^a f(x) dx, & f(2a - x) = f(x) \\ 0, & f(2a - x) = -f(x) \end{cases}$

Chapter 4

Unit 4: Multivariate Differentiation

Euler's Theorem on Homogeneous Functions

If $u(x, y)$ is homogeneous of degree n :

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu \quad (4.1)$$

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = n(n-1)u \quad (4.2)$$

2D Extrema & Lagrange Multipliers

- Critical points: $f_x = 0$, $f_y = 0$.
- Discriminant $D = rt - s^2 = f_{xx}f_{yy} - (f_{xy})^2$:
 - $D > 0, r > 0 \implies$ **Local Minimum**
 - $D > 0, r < 0 \implies$ **Local Maximum**
 - $D < 0 \implies$ **Saddle Point**
 - $D = 0 \implies$ **Inconclusive**
- Lagrange Multipliers: $\nabla f = \lambda \nabla g$.

Chapter 5

Unit 5: Multivariable Integration and Applications

Coordinate Transformations & Jacobians

- **Polar:** $x = r \cos \theta, y = r \sin \theta \implies dA = r \, dr \, d\theta$.
- **Cylindrical:** $x = r \cos \theta, y = r \sin \theta, z = z \implies dV = r \, dr \, d\theta \, dz$.
- **Spherical:** $x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi \implies dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$.
- **Planar Area:** $A = \iint_R 1 \, dA$.
- **Solid Volume:** $V = \iiint_{\Omega} 1 \, dV$.

Chapter 6

Unit 6: Introduction to Fourier Series

Euler Formulae on $(-L, L)$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right] \quad (6.1)$$

$$a_0 = \frac{1}{L} \int_{-L}^L f(x) dx \quad (6.2)$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx \quad (6.3)$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx \quad (6.4)$$

Symmetry & Half-Range Formulae

- **Even Function:** $b_n = 0$, $a_0 = \frac{2}{L} \int_0^L f(x) dx$, $a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$.
- **Odd Function:** $a_0 = a_n = 0$, $b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$.
- **Parseval's Identity:** $\frac{1}{L} \int_{-L}^L [f(x)]^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} (a_n^2 + b_n^2)$.

Chapter 7

High-Yield Examination Cheatsheet

7.1 Master Quick Reference Table

Table 7.1: MTH165 Complete High-Yield Summary

Unit	Core Topic	Essential Formula / Criterion
Unit 1	Matrix Rank & Echelon Eigenvalues & Cayley–Ham	$\rho(A) = \# \text{ pivots in REF}; \quad \rho(A) = \rho([A b]) \implies \text{Consistent}$ $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0; \quad A^3 - S_1A^2 + S_2A - S_3I = O$
Unit 2	Leibniz Product Rule Mean Value Theorems	$(uv)_n = \sum_{k=0}^n \binom{n}{k} u_{n-k} v_k$ Rolle: $f'(c) = 0$; LMVT: $f'(c) = \frac{f(b)-f(a)}{b-a}$; CMVT: $\frac{f'(c)}{g'(c)} = \frac{\Delta f}{\Delta g}$
Unit 3	King’s Definite Property Symmetry Properties	$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$ Even: $2 \int_0^a f(x)dx$; Odd: 0
Unit 4	Euler’s Homogeneous Thm 2D Extrema Discriminant	$xu_x + yu_y = nu; \quad x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$ $D = f_{xx}f_{yy} - (f_{xy})^2 > 0, f_{xx} > 0 \implies \text{Min}; D < 0 \implies \text{Saddle}$
Unit 5	Jacobians & Coordinates Area & Volume	Polar: $rdrd\theta$; Cyl: $rdrd\theta dz$; Spherical: $\rho^2 \sin \phi d\rho d\phi d\theta$ $A = \iint_R 1dA; \quad V = \iiint_\Omega 1dV$
Unit 6	Fourier Trigonometric Jump Discontinuity	$a_0 = \frac{1}{L} \int_{-L}^L f dx; \quad a_n = \frac{1}{L} \int_{-L}^L f \cos \frac{n\pi x}{L} dx; \quad b_n = \frac{1}{L} \int_{-L}^L f \sin \frac{n\pi x}{L} dx$ $S(x_0) = \frac{f(x_0^+) + f(x_0^-)}{2}$